

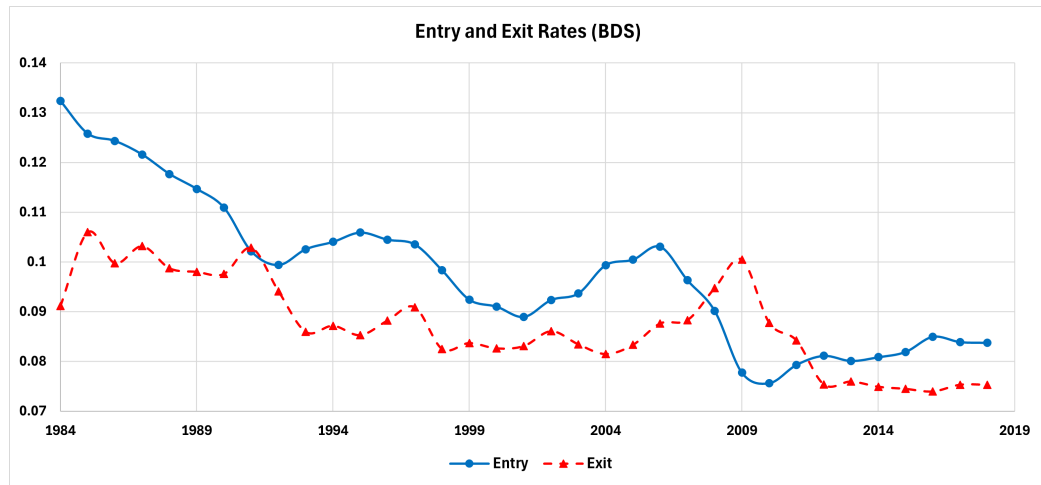
Firm Entry, Size, and Recession Recovery

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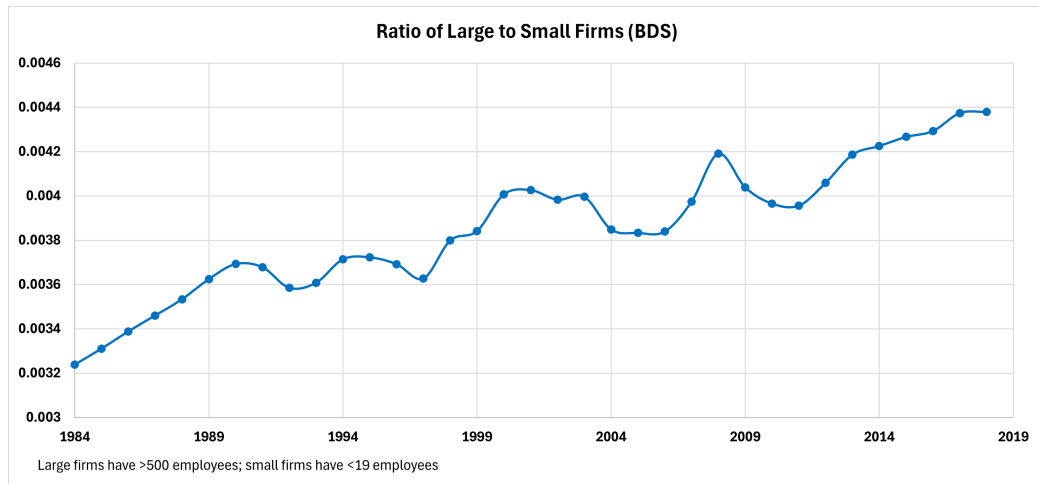
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Fall in Entry Rates



Entry and exit rates have fallen; exit rates relatively more stable

Increasing Concentration of Large Firms



Largest < 1% to smallest 99% by employment

Documented Trends & Economic Implications

Gutiérrez et al. (2021) use a NK model to show how entry costs can lead to:

- Fall in entry rates
- Increasing concentration across all sectors

At the same time, recession recoveries have been weaker (Galí et al., 2012)

Clementi et al. (2015) thru their model highlight opposing effects of entry costs on recession:

- Missing generation effect
- Selection effect

Do increased entry costs exacerbate recessions?

Agenda

General equilibrium model that accommodates firm entry & exit

- Random entry and continuation costs

Calibrate to cross sectional firm moments from BDS including:

- Unconditional size distribution
- Age-size moments

Examine response to TFP shock in various settings:

- Baseline (calibrated) with entry rate from 1990 (0.111)
- High entry cost environment with entry rate from 2003 (0.094)

Model Overview

Agents:

1. Competitive firms: DRS production with capital and labor, adjustment costs for capital
Incumbents: produce, decide whether to continue based on continuation cost
Potential entrants: decide whether to enter and produce based on entry cost
2. Representative household: Owns firms, supplies labor

Aggregate state: $(z, \mu) = \Theta$

- Exogenous aggregate TFP, z
- Distribution of firms, μ

Representative Household

- Period utility $U(C, 1 - N)$, and discount factor $\beta \in (0, 1)$
- Supplies: labor for wage $w(\Theta)$
- Equilibrium decision rules $C = C(\Theta)$, $N = N(\Theta)$
- Implied restrictions for equilibrium prices:
 - $w(\Theta) = \frac{D_2 U(C, 1 - N)}{D_1 U(C, 1 - N)}$
 - $d_m(\Theta_l) = \beta \frac{D_1 U(C'_m, 1 - N'_m)}{D_1 U(C, 1 - N)}$

► HH Problem

Firms - Incumbent

Firm's state (k, ε) : capital, idiosyncratic productivity

- Production technology: $z\varepsilon F(k, n)$
 - Enter period with ε ; retain with probability ρ
 - Probability $(1 - \rho)$ draw new ε from Pareto distribution: $G(\varepsilon)$; support $[\varepsilon_l, \varepsilon_h]$
- Post-production continuation cost: ξ
 - Draw $\xi \in [0, \bar{\xi}]$ from $H(\xi)$
- If continuing, invest subject to adjustment costs:
 - Adjustment cost: $q(k' - (1 - \delta)k)^2$
 - $k' = (1 - \delta)k + i$

Incumbent Firm Problem

Start of Period Value:

$$v_0(k, \varepsilon_L; \Theta_I) = \rho v_1(k, \varepsilon_L; \Theta_I) + (1 - \rho) \sum_{j=1}^{N_\varepsilon} \pi_j^\varepsilon v_1(k, \varepsilon_j; \Theta_I)$$

Production Value:

$$v_1(k, \varepsilon; \Theta_I) = \max_n \left[z\varepsilon F(k, n) - w(\Theta_I)n + (1 - \delta)k + \int_0^{\bar{\xi}} v_2(k, \xi, \varepsilon; \Theta_I) H(d\xi) \right]$$

where:

$$v_2(k, \xi, \varepsilon; \Theta_I) = \max \left\{ -q((1 - \delta)k)^2, \right. \\ \left. \max_{k'} \left[-(\xi + k' + q(k' - (1 - \delta)k)^2) + \sum_{m=1}^{N_\Theta} \pi_{I,m}^\Theta d_m(\Theta_I) v_0(k', \varepsilon; \Theta'_m) \right] \right\}$$

Continuation Decision Threshold

$\xi^T(k, \varepsilon; \Theta)$ is the cost where firm (k, ε) is indifferent between continuing or exiting:

$$\underbrace{-q((1-\delta)k)^2}_{\text{uninstall } k \text{ if exiting}} = -\xi^T(k, \varepsilon; \Theta) - \underbrace{(k' + q(k' - (1-\delta)k)^2)}_{k' \text{ adjustment cost}} + \underbrace{\frac{\beta}{p(\Theta)} \sum_{m=1}^{N_s} \pi_{l,m}^s V_0(k', \varepsilon; \Theta'_m)}_{\text{next period expected value}}$$

where: $\xi^T(k, \varepsilon; \Theta) \in [0, \bar{\xi}]$ and $V_0 \equiv p(\Theta)v_0(k, \varepsilon; \Theta)$

Potential Entrants

Constant, per period mass of potential entrants: M_e

- Endowed with k_e
- Draws entry cost, $\xi_e \in [0, \bar{\xi}_e]$ from $H_e(\xi_e)$
- Draws initial productivity, s from $G_e(s)$; support $[s_l, s_h]$, where $s_l = \varepsilon_l$ and $s_h = \frac{\varepsilon_h + \varepsilon_l}{2}$

At start of period, decides whether to enter *and produce* this period:

$$v_0^E(k_e, s; \Theta_l) = \max\{0, -\xi_e + v_1^E(k_e, s; \Theta_l)\}$$

- Probability $(1 - \rho)$ draw ε' from *incumbent* distribution, else retains s

Potential Entrant Problem

$$v_0^E(k, s; \Theta_I) = \max \left\{ 0, -\xi_e + v_1^E(k_e, s; \Theta_I) \right\}, \text{ where :}$$

$$v_1^E(k_e, s; \Theta_I) = \max_n \left[z\varepsilon F(k_e, n) - w(\Theta_I)n + (1 - \delta)k_e + \int_0^{\bar{\xi}} v_2^E(k_e, \xi, s; \Theta_I) H(d\xi) \right]$$

$$v_2^E(k_e, \xi, s; \Theta_I) = \max \left\{ -q((1 - \delta)k_e)^2, \max_{k'} \left[-(\xi + k' + q(k' - (1 - \delta)k_e)^2) \right. \right. \\ \left. \left. + \sum_{m=1}^{N_\Theta} \pi_{l,m}^\Theta d_m(\Theta_I) \left(\rho v_1^E(k', s; \Theta'_m) + (1 - \rho) \sum_{j=1}^{N_\varepsilon} \pi_j^\varepsilon v_1(k', \varepsilon_j; \Theta'_m) \right) \right] \right\}$$

Threshold entry cost:

$$\xi_e^T(k_e, s; \Theta) = v_1^E(k_e, s; \Theta)$$

Calibration: Functional Forms & Parameters

- Annual frequency
- $z\varepsilon F(k, n) = z\varepsilon k^\alpha n^\nu, \alpha + \nu < 1$
- $\xi \sim U(0, \bar{\xi}), \xi_e \sim U(0, 1)$
- $s \sim U(s_l, s_h)$

$$U(C, 1 - N) = \ln(C) + \psi(1 - N)$$

$$k_e = \chi[\int_\mu k \mu(d[k \times \varepsilon]) + k_e M_e]$$

$$M_e = 0.131 \rightarrow \text{prod. mass} = 1$$

	Parameter		Data		Model
β	discount factor	0.961	real interest rate	0.040	0.040
ψ	leisure preference	2.620	labor hours	0.333	0.335
ν	labor share	0.600	labor share	0.600	0.600
δ	depreciation	0.069	investment/capital	0.069	0.069
α	capital share	0.260	capital/output	2.250	2.109

Firm Level Parameters: Jointly Calibrated

Business Dynamics Statistics (1990-2003)

ε_l	ε_h	ρ	$\bar{\xi}$	χ	q
0.400	2.060	0.740	0.1708	0.3517	0.0033

Unconditional size distribution:

Employment Bins	Emp. Share	Pop. Share	
		Data	Model
Small (1-19)	0.202	0.886	0.865
Med. (20-499)	0.319	0.111	0.104
Large (500+)	0.479	0.003	0.031

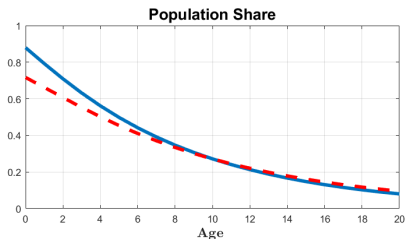
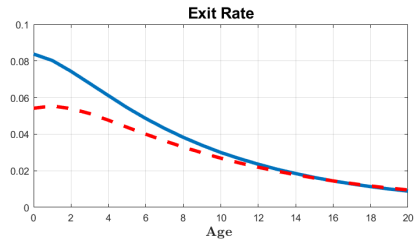
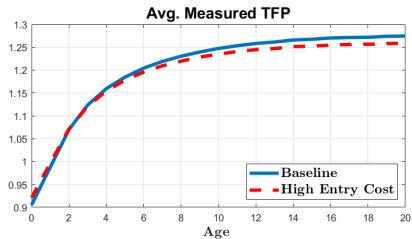
Age-size distribution:

Age	Data	Model
Entrant:	0.262	0.262
1 year:	0.343	0.343

Young firm mass: Data: 0.400 | Model: 0.513

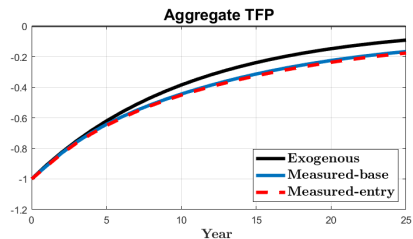
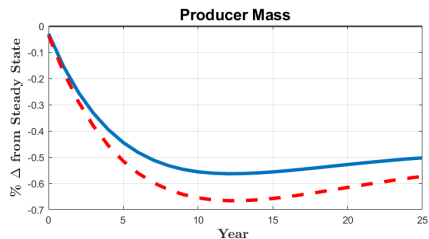
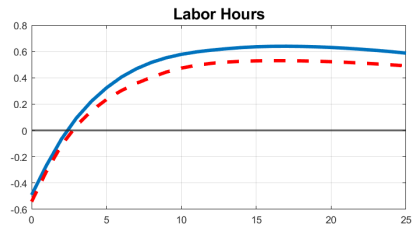
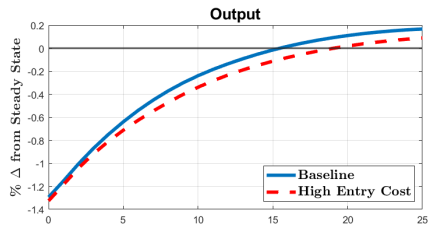
Relative Measures in Cohort's Life-cycle

1990 Entry Rate (0.111) vs 2003 Entry Rate (0.094)



Impulse Responses to 1% TFP Shock

Recovers at 0.091% per year



Thank You!

Still early wrt direction; any comments welcome!

Appendix and Further Equations

Market Clearing

Goods Market Clearing: $C(\Theta) = Y(\Theta) - I(\Theta) - \Phi(\Theta) - \Phi_e(\Theta) - \Phi_k(\Theta)$, where:

$$K(\Theta) = \int_{\mu} k \mu(d[k \times \varepsilon]) + k_e M_e$$

$$N(\Theta) = \rho \int_{\mu} n(k, \varepsilon_L; \Theta) \mu(d[k \times \varepsilon_L]) + (1 - \rho) \int_{\mu} n(k, \varepsilon; \Theta) \mu(d[k \times \varepsilon])$$

$$+ M_e \sum_{i=1}^{N_s} \pi_i^s \int_0^{\xi_e^T(k_e, s_i; \Theta)} n(k_e, s_i; \Theta) H_e(d\xi_e)$$

$$Y(\Theta) = \rho \int_{\mu} z \varepsilon F(k, n(k, \varepsilon_L); \Theta) \mu(d[k \times \varepsilon_L]) + (1 - \rho) \int_{\mu} z \varepsilon F(k, n(k, \varepsilon); \Theta) \mu(d[k \times \varepsilon])$$

$$+ M_e \sum_{i=1}^{N_s} \pi_i^s \int_0^{\xi_e^T(k_e, s_i; \Theta)} z \varepsilon F(k_e, n(k_e, s_i); \Theta) H_e(d\xi_e)$$

Continuation, Entry & Capital Adjustment Costs

$$\Phi(\Theta) = \rho \int_{\mu} \int_0^{\xi^T(k, \varepsilon_L; \Theta)} \xi H(d\xi) \mu(d[k \times \varepsilon_L]) + (1 - \rho) \int_{\mu} \int_0^{\xi^T(k, \varepsilon; \Theta)} \xi H(d\xi) \mu(d[k \times \varepsilon])$$

$$+ M_e \sum_{i=1}^{N_s} \pi_i^s \int_0^{\xi_e^T(k_e, s_i; \Theta)} \int_0^{\xi^T(k_e, s_i; \Theta)} \xi H(d\xi) H_e(d\xi_e)$$

$$\Phi_e(\Theta) = M_e \sum_{i=1}^{N_s} \pi_i^s \int_0^{\xi_e^T(k_e, s_i; \Theta)} \xi_e H_e(d\xi_e)$$

$$\Phi_k(\Theta) = \rho \int_{\mu} \left[\int_0^{\xi^T(k, \varepsilon_L; \Theta)} q[k'(k, \varepsilon_L; \Theta) - (1 - \delta)k]^2 H(d\xi) + \int_{\xi^T(k, \varepsilon_L; \Theta)}^{\bar{\xi}} q[(1 - \delta)k]^2 H(d\xi) \right] \mu(d[k \times \varepsilon_L])$$

$$+ (1 - \rho) \int_{\mu} \left[\int_0^{\xi^T(k, \varepsilon; \Theta)} q[k'(k, \varepsilon; \Theta) - (1 - \delta)k]^2 H(d\xi) + \int_{\xi^T(k, \varepsilon; \Theta)}^{\bar{\xi}} q[(1 - \delta)k]^2 H(d\xi) \right] \mu(d[k \times \varepsilon])$$

$$+ M_e \sum_{i=1}^{N_s} \pi_i^s \int_0^{\xi_e^T(k_e, s_i; \Theta)} \int_0^{\xi^T(k_e, s_i; \Theta)} q[k'(k_e, s_i; \Theta) - (1 - \delta)k_e]^2 H(d\xi) H_e(d\xi_e)$$

Investment

$$I(\Theta) =$$

$$\begin{aligned} & \rho \int_{\mu} \left[\int_0^{\xi^T(k, \varepsilon_L; \Theta)} [k'(k, \varepsilon_L; \Theta) - (1 - \delta)k] H(d\xi) - \int_{\xi^T(k, \varepsilon_L; \Theta)}^{\bar{\xi}} (1 - \delta)k H(d\xi) \right] \mu(d[k \times \varepsilon_L]) \\ & + (1 - \rho) \int_{\mu} \left[\int_0^{\xi^T(k, \varepsilon; \Theta)} [k'(k, \varepsilon; \Theta) - (1 - \delta)k] H(d\xi) - \int_{\xi^T(k, \varepsilon; \Theta)}^{\bar{\xi}} (1 - \delta)k H(d\xi) \right] \mu(d[k \times \varepsilon]) \\ & + M_e k_e + M_e \sum_{i=1}^{N_s} \pi_i^s \int_0^{\xi_e^T(k_e, s_i; \Theta)} \int_0^{\xi^T(k_e, s_i; \Theta)} [k'(k_e, s_i; \Theta) - (1 - \delta)k_e] H(d\xi) H_e(d\xi_e) \\ & - M_e k_e \sum_{i=1}^{N_s} \pi_i^s \left[\int_0^{\xi_e^T(k_e, s_i; \Theta)} \int_{\xi^T(k_e, s_i; \Theta)}^{\bar{\xi}} H(d\xi) H_e(d\xi_e) + \int_{\xi_e^T(k_e, s_i; \Theta)}^{\bar{\xi}_e} H_e(d\xi_e) \right] \end{aligned}$$

Household Problem

Representative HH maximizes lifetime value, $W(\lambda; \Theta)$

- Chooses: consumption, c , labor, n^h , shares, λ'

$$W(\lambda; \Theta) = \max_{c, n^h, \lambda'} \left[U(c, n^h) + \beta W(\lambda'; \Theta'_m) \right]$$

subject to:

$$\begin{aligned} c + \int \rho_1(k', \varepsilon'; \Theta'_m) \lambda'(d[k' \times \varepsilon']) + \int \rho_1(k', s'; \Theta'_m) \lambda'(d[k' \times s']) \leq \\ w(\Theta) n^h + \int \rho_0(k, \varepsilon; \Theta_I) \lambda(d[k \times \varepsilon]) + \int \rho_0(k, \varepsilon; \Theta_I) \lambda(d[k \times \varepsilon]) \\ \mu' = \Gamma(\Theta_I) \end{aligned}$$

- ρ_1 is ex-dividend price of a share, ρ_0 is dividend-inclusive value of a share, $\Gamma(\mu)$ is known

Steady State in Two Economies

Study effects of TFP shock under different conditions

- Baseline
- Increase mean, dispersion ξ_e such that entry rate falls to 9.37%

	Y	C	I	Mass	L:S ratio	$\frac{\Phi_c}{Y}$	$\frac{\Phi_e}{Y}$	$\frac{Q}{Y}$
Baseline	0.668	0.456	0.096	1	0.036	0.094	0.063	0.016
High Entry Cost	0.653	0.455	0.094	0.963	0.037	0.097	0.047	0.017

► back

Changes in Entry, Exit, and Wage

